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ESG's path to value: the role of total factor productivity and R&D innovation investment

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ABSTRACT

This study examines the effect of environmental, social, and governance (ESG) performance on company value, focusing on the mediating role of total factor productivity (TFP) and the moderating role of R&D innovation investment. Using data from Chinese A-share companies (2012–2022) and a moderated mediation model, the study found that ESG performance enhances company value both directly and indirectly through TFP. R&D innovation investment strengthens both effects, highlighting the synergy between ESG and innovation strategies in maximizing value creation. This effect is most pronounced in the high-tech and manufacturing sectors, underscoring the importance of integrating ESG initiatives with innovation to foster sustainable growth.

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ESG performance; company value; total factor productivity; R&D innovation investment; moderated mediation analysis

1. Introduction

The climate risks associated with global warming extend beyond environmental concerns, presenting worldwide challenges in economic and social domains that demand immediate and collaborative action. Leading nations, including China, have initiated carbon neutrality endeavors, such as China's '30–60 goal,' which aims for 'carbon peak' and 'carbon neutrality' by 2030 and 2060, respectively. In this context, companies, viewed as micro entities, are required to extend their responsibilities for carbon emission reduction to the micro-company level to achieve the overarching goal of 'dual carbon' (Doda et al. 2016).

Environmental, social, and governance (ESG) factors are key tools for achieving carbon neutrality (Chen, Song, and Gao 2023). ESG assessments serve as quantifiable indicators for evaluating corporate sustainability and social performance (Clark and Viehs 2014). This approach helps quantify the impact of social responsibility initiatives on company value. Empirical evidence shows that investors, stakeholders, and the public increasingly use ESG standards to assess corporate capabilities in environmental impact, social responsibility, and governance (Senadheera et al. 2022). Consequently, ESG performance has become a significant benchmark for evaluating corporate environmental and social responsibility, with a clear impact on company value (Shakil 2021).

ESG is crucial for economic sustainability and high-quality development (Sun et al. 2022). It is closely linked to R&D innovation (Y. Li and Li 2024). China's 'dual carbon'

strategy seeks to advance ecological civilization and low-carbon growth while boosting productivity, TFP, and production capabilities, with R&D innovation playing a key role. Both ESG performance and R&D innovation investment strengthen core competitiveness and drive sustainable growth. Understanding their combined impact on company value is, therefore, essential.

Previous studies show that the effect of ESG performance on company value is complex and dynamic. The effects, mediating pathways, and moderating factors of this relationship continue to evolve. First, regarding causality, the findings on the correlation between ESG performance and company value remain controversial (Au et al. 2023; Chen, Song, and Gao 2023; T. Li et al. 2021). However, most studies suggest a positive effect of ESG performance on company value (Aydoğmuş, Gülay, and Ergun 2022; Broadstock et al. 2021; Jin and Lei 2023; Shen et al. 2023). Conversely, some researchers have identified a negative correlation (Junius et al. 2020; Palupi 2023; Xin, Zhang, and Xiang 2024).

Secondly, in exploring mediating pathways, scholars have elucidated various mechanisms through which ESG performance influences company value. Research suggests that ESG performance enhances company value by mitigating information asymmetry, reducing financing costs, improving profitability, lowering operational risks, and increasing productivity (Shen et al. 2023; N. Wang et al. 2023). Total Factor Productivity (TFP), as a key measure of corporate productivity and an important driver of company value, has been central to these discussions. Studies have shown that strong ESG performance correlates with increased TFP (Deng, Li, and Ren 2023; Ma, Gao, and Sun 2022; X. Yu and Chen 2024). Further research highlights the mediating role of TFP, suggesting that ESG performance positively affects company value both directly and indirectly through TFP.

Thirdly, scholars have investigated moderating factors affecting the relationship between ESG performance and company value, emphasizing variables that may enhance or diminish this effect. For example, Boulhaga et al. (2023) reported that higher internal control quality strengthens this effect. Fu and Li (2023) demonstrated that higher levels of digital transformation further intensify ESG's influence on company value. Despite these contributions, significant gaps remain in understanding the underlying mechanisms influencing this relationship.

A review of the literature shows that few studies have explored R&D innovation investment as a moderating factor in the ESG-company value relationship, as noted by Bhattacharya, Okafor, and Pradeep (2021). Duan, Yang, and Xiong (2023) emphasized the importance of investigating how R&D innovation investment interacts with ESG performance to affect company value as an aspect often overlooked in ESG research. Their findings indicate that R&D innovation investment may negatively moderate the effect of ESG performance on company value, particularly in the manufacturing industry, a conclusion that warrants further empirical validation. Whether this moderating role is a 'crowding out effect' or a 'synergistic effect' remains unclear. Therefore, it is crucial to explore in depth how R&D innovation investment interacts with ESG performance to influence company value, underscoring the need for comprehensive analysis of its moderating role within the ESG framework.

This identifies a critical research gap which reflects the need to integrate the mediating effect of TFP and the moderating role of R&D innovation investment into a unified framework, as well as explore their interactions. To address this gap, this study formulates a moderated mediation model centered on ESG performance, exploring whether ESG

performance affects company value via TFP and whether R&D innovation investment influences these effects.

The primary contributions of this study are as follows: First, this article provides empirical evidence that ESG performance positively affects company value through TFP. Second, the research reveals a 'synergistic effect' between R&D innovation investment and ESG initiatives, demonstrating that R&D innovation investment amplifies the impact of ESG performance on company value. Third, the study distinguishes between high-tech and non-high-tech industries, showing that the positive effect of R&D innovation investment is more pronounced in high-tech companies, which enhances the specificity and applicability of the findings. Fourth, the study finds that the effect of ESG performance on company value differs between manufacturing and non-manufacturing industries, with manufacturing companies benefiting more from integrating ESG performance with R&D innovation investment, thereby generating value-added synergies. Importantly, the research shows that R&D innovation investment negatively moderates the impact of ESG on TFP but positively moderates the effect of TFP on company value. The results suggest that improving R&D innovation investment magnifies the indirect effect of ESG performance on company value via TFP. These insights provide valuable guidance for strategic decisions regarding long-term value investments, such as ESG and R&D innovation.

The findings of this study offer significant implications for companies, investors, and governments seeking a deeper understanding of green development. The results elucidate how ESG practices enhance company value, providing management with new insights into improving ESG performance, advancing green transformation, and increasing R&D innovation investment. For investors, the study establishes a benchmark for evaluating corporate sustainability and long-term value, supporting effective resource allocation and value discovery. For governments, the evidence underscores the impact of policies on green, low-carbon, and innovation-driven development, as well as TFP improvement, serving as a reference for future policy formulation.

2. Literature review and theoretical framework

2.1. ESG performance and company value

The efficacy of ESG performance in enhancing company value remains controversial (Pu, 2022; N. Wang et al. 2023). Companies engaging in ESG practices navigate a complex interplay of stakeholder demands, balancing the 'cost effect' with the 'value effect.' According to stakeholder theory, fulfilling social responsibilities generates mutual benefits. ESG provides a framework for meeting stakeholder expectations, and companies that excel in these areas tend to outperform less responsible peers (Li et al. 2021). Signalling theory (Spence 2002) emphasizes the role of information transparency, with ESG acting as a signal of a company's environmental, social, and governance commitment, conveying long-term value to investors (Ren and Cheng 2024). Similarly, reputation theory further emphasizes ESG's role in enhancing corporate reputation, a critical intangible asset that drives competitiveness (Walker 2010). Consequently, strong ESG performance often elicits a favorable response in terms of company value.

A growing body of research suggests that ESG performance positively impacts company value (Chen, Song, and Gao 2023; Cheng, Kim, and Ryu 2023; Habib and Mourad 2024; Sinha Ray and Goel 2022; Veeravel, Sadharma, and Kamaiah 2024; Wu et al. 2022; X. Yu and Xiao 2022; G. Zhou, Liu, and Luo 2022). Friede, Busch, and Bassen (2015) reported that approximately 90% of studies found a non-negative relationship between ESG performance and corporate financial performance (CFP). However, not all findings are consistent. Pu (2023) identified an inverted U-shaped effect of ESG activities on firm performance, suggesting that excessively high ESG levels may lead to diminishing returns. Palupi (2023) argued that ESG reporting could incur ecological costs and create agency conflicts, potentially harming financial outcomes. Moreover, Junius et al. (2020) and Xin, Zhang, and Xiang (2024) found that ESG scores do not consistently influence company performance or market value. These discrepancies are likely attributable to differences in variable measurement, regression techniques, and institutional contexts (Wang et al., 2023). Based on this, hypothesis 1 is proposed as follows:

H1: ESG performance has a positive effect on company value.

2.2. ESG performance, total factor productivity, and company value

TFP is a key indicator of corporate productivity and an important determinant of company value (Tian and Twite 2011). Resource dependency theory offers the theoretical basis for TFP's mediating role, proposing that effective management of external resources, including social and environmental factors, enhances productivity and creates value. ESG performance is believed to positively influence TFP by promoting efficient resource use, fostering innovation, and enhancing employee engagement, which are critical for improving productivity (Ding, Han, and Wang 2024; X. Yu and Chen 2024). ESG initiatives streamline processes, encourage innovation, and reduce inefficiencies, all of which are expected to enhance TFP (M. Wang et al. 2021; P. Yu, Zuo, and Lian 2024). Improved TFP subsequently drives company value by increasing profitability, strengthening competitiveness, and attracting investors. Thus, ESG practices act as a mechanism to enhance TFP and, elevate company value.

Empirical evidence supports this theoretical perspective. Studies consistently demonstrate a positive effect of ESG performance on TFP (Deng, Li, and Ren 2023; Ge et al. 2022; Ma, Gao, and Sun 2022). Ma, Gao, and Sun (2022) showed that ESG alleviates financial constraints and stimulates investment in innovation, thereby contributing to TFP growth. Deng, Li, and Ren (2023) found that an increase in corporate ESG ratings correlates with a 1.19% rise in TFP, driven by reduced financial constraints, an enhanced public reputation, and improved human resource efficiency. Tunio et al. (2021) and further emphasized ESG's role in improving employee productivity and optimizing resource allocation, ultimately boosting company value. Firms with strong ESG scores demonstrate better resource utilization, higher innovation, and more effective employee management, all of which contribute to increased TFP (Niu and Wang 2024). This elevated TFP enhances company value by increasing profitability, strengthening competitiveness, and boosting investor confidence (Rahmatulloh and Suranta 2023). Thus, good ESG performance

promotes optimal resource allocation within the company and improves productivity, which enhances company value. Based on this, hypothesis 2 is proposed.

H2: Total factor productivity is a mediating path through which ESG performance enhances company value.

2.3. Moderating effect of R&D innovation investment

ESG and R&D activities play a vital role in enhancing company performance and sustainability. Research shows that ESG investments and technological innovation are mutually reinforcing, creating a 'synergistic effect' (Alam et al. 2019; Yang et al. 2024). There is a strong correlation between ESG performance, R&D innovation investment, and corporate outcomes. Companies with higher R&D and patent investments often demonstrate stronger ESG performance (Dicuonzo et al. 2022). R&D innovation investment positively influences green innovation, while ESG practices drive an increase in green invention patents, both of which contribute to company value (Xu, Liu, and Shang 2021).

Existing research indicates that R&D innovation investment positively moderates environmental and social performance, ultimately improving financial outcomes (Baek and Lee 2024; X. Zhou, Wu, and Huang 2024). This study proposes that R&D innovation investment moderates the relationship between ESG performance and company value. Firstly, high R&D innovation investment can drive environmentally friendly, socially responsible, and effective governance practices. Secondly, R&D innovation investment is a key component of ESG factors. Thirdly, the level of R&D innovation investment may influence the pathway through which ESG performance affects company value. These findings highlight the potential of ESG performance and R&D innovation investment to enhance company value. Based on this, hypothesis 3 is proposed as follows:

H3: R&D innovation investment moderates the effect of ESG performance on company value.

2.4. Moderated mediation effect of R&D innovation investment

ESG performance enhances company value by improving TFP, an effect that is further amplified by R&D innovation investment. Integrating ESG initiatives with R&D innovation fosters sustainable practices, resource efficiency, and adaptability which are the key drivers of TFP and company value. Innovation theory emphasizes that innovation is essential for economic growth (Kasongo and Makamu 2024). High R&D innovation investment enhances the effectiveness of ESG practices by promoting technological progress, optimizing resource allocation, and increasing market adaptability (Di Simone, Petracci, and Piva 2022; Lian, Li, and Cao 2023; Wang et al., 2023; Xu, Liu, and Shang 2021). Thus, companies that align ESG initiatives with R&D innovation investment can significantly amplify the impact of TFP on company value.

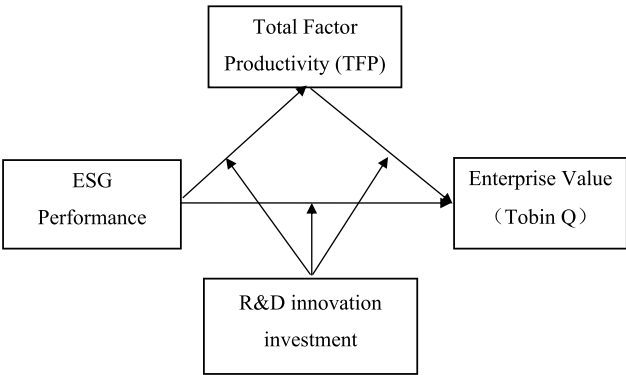


Figure 1. Moderated mediation model showing the effect of ESG performance on company value.

Resource allocation theory suggests that prioritizing R&D innovation investment over ESG may lead to short-term trade-offs, as resource allocation often operates as a zero-sum game (Duan, Yang, and Xiong 2023). Increasing R&D innovation investment could limit resources available for ESG initiatives, potentially weakening their immediate impact on TFP. However, TFP gains driven by R&D innovation investment significantly enhance productivity and investor confidence, thereby boosting company value (Ding, Han, and Wang 2024). R&D innovation investment is, therefore, not only a catalyst for innovation but also a supporting mechanism that strengthens the positive impact of ESG performance on company value by improving resource utilization and fostering market-oriented innovation (Zheng and Feng 2024).

Empirical evidence supports this relationship. Duan, Yang, and Xiong (2023) found that while ESG performance generally enhances company value, high R&D intensity can negatively moderate this effect in the short term, particularly in resource-constrained industries such as manufacturing. In contrast, Kong et al. (2023) demonstrated that technological innovation strengthens the positive effect of ESG performance on company value, highlighting the synergy between R&D innovation investment and ESG initiatives. These findings suggest that, although high R&D innovation investment may initially divert resources from ESG efforts, the long-term productivity and market gains from enhanced innovation ultimately amplify company value. Based on this, hypothesis 4 is proposed as follows:

H4: The mediation effect is moderated by R&D innovation investment and the conditional indirect effect is stronger under conditions of higher R&D innovation investment.

Figure 1 presents moderated mediation model showing the effect of ESG performance on company value in this study.

3. Research design

3.1. Data source and sample selection

This study used Chinese A-share listed companies from 2012 to 2022 as the research sample. To ensure representativeness and scientific rigor, the sample was processed as

follows: (1) excluding delisted companies and those marked as ST or ST* due to delisting risks; (2) excluding companies in the financial industry; and (3) excluding observations with missing data on ESG ratings, Tobin's Q, or other relevant variables. To mitigate the impact of outliers, continuous variables were Winsorized at the 1% and 99% levels. After processing, the final sample comprised 4,389 listed companies, resulting in an unbalanced panel dataset with 31,310 annual observations from 2012 to 2022. The data were sourced from the China Stock Market & Accounting Research (CSMAR) and Wind databases.

3.2. Definition of variables

3.2.1. Explained variable: company value

Most scholars (Abdi, Li, and Càmara-Turull 2022; Duan, Yang, and Xiong 2023; X. Yu and Xiao 2022; Zhang 2022; G. Zhou, Liu, and Luo 2022) have adopted Tobin's Q, a market-based performance measure, to assess company value. This study followed this approach by using Tobin's Q as a proxy for company value. Tobin's Q was calculated as the sum of the year-end market value of outstanding shares, non-outstanding shares, and net liabilities, divided by the year-end total assets. Additionally, a robustness test was conducted using business market value (MV) as an alternative proxy variable.

3.2.2. Explanatory variable: ESG performance

Most studies have used ESG ratings to evaluate corporate ESG performance. ESG ratings from Bloomberg, Huazheng, and SynTao Green Finance are widely recognized and applied in both domestic and international research (Eliwa, Aboud, and Saleh 2021). Among these, the most commonly used sources are Huazheng (29%), Bloomberg (24%), and SynTao Green Finance (12%) (Shen et al. 2023). This study characterized ESG performance using Huazheng ESG ratings, following previous research practices. Huazheng's ESG evaluation system is divided into nine grades. This study assigned scores accordingly, with AAA given a score of 9 and C given a score of 1, where higher scores indicate better ESG performance. Annual ESG performance was calculated by averaging the quarterly evaluation scores.

3.2.3. Mediating variable: Total Factor Productivity (TFP)

TFP measures the level of economic and social value created by a company. Various methods exist for measuring TFP, including the Cobb-Douglas production function, the Olley and Pakes (OP) method, and the Levinsohn and Petrin (LP) method. The LP method addresses limitations in the OP method, particularly regarding missing proxy variable estimates. Thus, following Ma, Gao, and Sun (2022), this study employed the LP method to measure TFP and used the OLS method for robustness testing, as recommended by Levinsohn and Petrin (2003).

3.2.4. Moderating variable: R&D innovation investment

R&D innovation investment is pivotal for driving innovation outputs and is closely linked to company capabilities and performance. Its magnitude reflects the level of specialization in a technological domain and serves as a key driver of technological progress. Key indicators include total R&D innovation investment, investment intensity, and R&D

personnel count. Following Rahman and Howlader (2022) and Zhang (2022), this study used aggregate R&D innovation investment as a measure. Due to the large scale of the data, a natural logarithmic transformation was applied for the regression analysis to ensure smoothness.

3.2.5. Control variables

Based on prior literature (Chen, Song, and Gao 2023; Jin and Lei 2023; Ma, Gao, and Sun 2022; Rohendi, Ghazali, and Ratmono 2024), this study selected control variables covering company characteristics, financial attributes, and governance features to enhance robustness and theoretical rigor. Specifically, company characteristics include company size (Size) and company age (Age) to assess their impact on TFP and company value. Company size affects resource allocation, economies of scale, and bargaining power, influencing TFP and company value, while company age represents accumulated experience, potentially affecting productivity and market valuation.

Financial attributes included company growth (Growth), asset-liability ratio (Lev), and operating cash flow (CFO). Company growth reflects future profit prospects, the asset-liability ratio measures capital structure and financial risk, and operating cash flow indicates financial health, each potentially impacting TFP and company value. Governance characteristics included the shareholding ratio of the largest shareholder (Top1), board independence (Indep), and institutional investor shareholding ratio (INST). The largest shareholder's ratio influences decision-making and agency problems, while board independence and institutional investor shareholding enhance governance efficiency and ESG performance.

Additionally, year, industry (Indu), and individual (Id) fixed effects were included to control for macroeconomic conditions, unobservable time-varying factors, and cross-industry heterogeneity. This comprehensive approach mitigates external interference, thereby improving the robustness and reliability of the research results. Table 1 provides the variable definitions.

3.3. Model design

Building upon prior work, the models of this study were designed to examine mediation effects, moderated effects, and combined mediation with moderation. Following Wen et al. (2004), a three-step method was used to test the mediation effect. To test the proposed hypotheses, the study adopted the model and methodology outlined by Kaiyue, Tianxu, and Lei (2023), and empirical research was conducted accordingly.

$$\text{TobinQ}_{i,t} = a_0 + a_1 \text{ESG}_{i,t} + a_j \sum_j \text{Controls}_{i,t} + \sum \text{Year} + \sum \text{Indu} + \sum \text{Id} + \varepsilon_{i,t} \quad (1)$$

$$\text{TFP}_{i,t} = b_0 + b_1 \text{ESG}_{i,t} + b_j \sum_j \text{Controls}_{i,t} + \sum \text{Year} + \sum \text{Indu} + \sum \text{Id} + \varepsilon_{i,t} \quad (2)$$

$$\text{TobinQ}_{i,t} = c_0 + c_1 \text{ESG}_{i,t} + c_2 \text{TFP}_{i,t} + c_j \sum_j \text{Controls}_{i,t} + \sum \text{Year} + \sum \text{Indu} + \sum \text{Id} + \varepsilon_{i,t} \quad (3)$$

Table 1. Defining variables.

Types	Variables	Symbol	Definitions
Dependent Variable	Company value	Tobin Q	(Market value of outstanding shares + Market value of non-outstanding shares + Book value of liabilities)/Total assets
Independent Variable	ESG performance	ESG	Based on Huazheng ESG rating from AAA to C, it is rated from 9 to 1. Annual evaluation values are calculated based on quarterly averages
Mediator Variable	Total Factor Productivity	TFP	The TFP of enterprises calculated through Levinsohn–Petrin (LP) method
Moderator Variable	R&D innovation investment	RDC	Corporate R&D innovation investment plus 1 takes the natural logarithm
Control Variables	Company size	Size	Natural logarithm of total assets
	Company age	Age	Natural logarithm from year of listing to year of observation
	Company growth	Growth	Main operating income growth rate
	Asset-Liability Ratio	Lev	Total liabilities/Average total assets
	Operating cash flow	CFO	Net cash flows from operating activities/Total assets
	Shareholding ratio of the largest shareholder	Top1	Number of shares held by the largest shareholder/Total number of shares
	Board independence	Indep	Ratio of number of sole directors to board size
	Institutional investor shareholding	INST	Total number of shares held by institutional investors/Total number of share capital

$$TobinQ_{i,t} = d_0 + d_1 ESG_{i,t} + d_2 RDC_{i,t} + d_3 ESG_{i,t} \times RDC_{i,t} + d_j \sum_j Controls_{i,t} + \sum Year + \sum Indu + \sum Id + \varepsilon_{i,t} \quad (4)$$

$$TFP_{i,t} = e_0 + e_1 ESG_{i,t} + e_2 RDC_{i,t} + e_3 ESG \times RDC_{i,t} + e_i \sum_j Controls_{i,t} + \sum Indu + \sum Id + \varepsilon_{i,t} \quad (5)$$

$$TobinQ_{i,t} = f_0 + f_1 ESG_{i,t} + f_2 RDC_{i,t} + f_3 ESG_{i,t} \times RDC_{i,t} + g_1 TFP_{i,t} + g_2 TFP_{i,t} \times RDC_{i,t} + f_j \sum_j Controls_{i,t} + \sum Year + \sum Indu + \sum Id + \varepsilon_{i,t} \quad (6)$$

The models in this study were developed to examine various effects: Model (1) tested the effect of ESG performance on company value; Model (2) tested the effect of TFP on company value; Model (3) tested the mediating role of TFP between ESG performance and company value; Model (4) tested the moderating role of R&D innovation investment on the ESG-company value relationship; and Models (5) and (6) tested the moderating effect of R&D innovation investment on the mediating role of TFP in the ESG-company value relationship. Control variables (Controls), year fixed effects (Year), industry fixed effects (Indu), individual fixed effects (Id), and random error terms ($\varepsilon_{i,t}$) were included to improve the robustness of the analysis.

A three-step mediation regression was conducted to assess whether TFP mediated the effect of ESG performance on company value. In Model 1, where the mediating variable (TFP) was not included, the coefficient for ESG performance (a_1) was significant, indicating a total effect of ESG on company value. If a_1 was not significant, further mediation analysis was stopped. In Models 2 and 3, TFP was included as the mediator. If coefficients b_1 and

c2 were significant, TFP was confirmed as a mediator. If c1 was also significant, TFP had a partial mediation effect; if not, TFP had a full mediation effect.

To mitigate covariance issues, the data for testing moderating effects were centered. The significance of interaction terms was analyzed to confirm the presence of moderating effects. Building on this premise, a moderating model of ESG performance and company value was used to explore the moderating role of R&D innovation investment. Subsequently, a mediation model with moderation was constructed to determine whether the mediating effect of TFP on the ESG-company value relationship was moderated by R&D innovation investment. If any of the coefficients for e1 and g2, e3 and g1, or e3 and g2 were significant, it would indicate that the mediating role of TFP was moderated by R&D innovation investment.

4. Empirical results and analysis

4.1. Descriptive statistics of the variables

The descriptive statistics of the variables are presented in Table 2. Tobin's Q had a mean value of 2.072 with a standard deviation of 1.370, ranging from a minimum of 0.850 to a maximum of 8.871, reflecting a substantial disparity in the market value of Chinese listed companies. The mean ESG performance was 4.126, with values ranging from 1.250 to 6.250, and a median of 4.000, suggesting that most Chinese listed companies achieved an average ESG rating within the B-BB range, indicating a moderate level of performance. However, there were significant variations in ESG performance among companies. For the mediating variable, TFP had a mean of 8.319, with values ranging from 6.090 to 11.12, highlighting notable differences in TFP across the sampled companies.

For the moderating variables, a notable discrepancy existed between the maximum and minimum values of enterprises' R&D investment, with a standard deviation of 6.237, indicating substantial variation among different enterprises. The control variables were largely consistent with findings from previous scholarly studies.

The distribution of the sample data was smooth, with no outliers, and showed differentiation. To assess multicollinearity, Table 2 presented the variance inflation factor (VIF) results. The VIF values ranged from 1 to 4, with most control variables below 2, all well

Table 2. Descriptive statistics of variables.

Variable	N	Mean	SD	Min	Median	Max	VIF
Tobin Q	31310	2.072	1.370	0.850	1.628	8.871	—
ESG	31310	4.126	1.057	1.250	4.000	6.250	1.20
TFP	31310	8.319	1.053	6.090	8.218	11.12	2.82
RDC	31310	15.63	6.237	0.000	17.75	21.82	1.14
Size	31310	22.27	1.298	19.90	22.08	26.28	3.36
Age	31310	2.210	0.782	0.693	2.303	3.367	1.47
Growth	31310	0.163	0.427	−0.579	0.096	2.710	1.03
Lev	31310	0.429	0.207	0.059	0.419	0.929	1.55
CFO	31310	0.048	0.068	−0.156	0.046	0.247	1.09
Top1	31310	33.66	14.73	8.448	31.29	73.82	1.42
Indep	31310	35.01	6.728	20.00	33.33	57.14	1.02
INST	31310	43.77	24.83	0.256	45.04	92.08	1.66

Source: Data processing results from Stata software.

under the threshold of 10, and an average VIF of 1.62. These results indicate no significant multicollinearity among the variables.

4.2. Hypothesis testing

4.2.1. ESG performance and company value

Table 3 presents the regression results for the effect of ESG performance on company value. This study accounted for fixed effects at the year, industry, and individual levels, and clustered standard errors at the individual level. This approach mitigated potential biases arising from unobserved heterogeneity and intra-individual correlation, enhancing the robustness of the findings. In column (1), the ESG coefficient was significantly positive at the 1% level, controlling only for company fundamentals and financial indicators. Column (2) introduced governance characteristics, including the shareholding proportion of the largest shareholder, the proportion of independent directors, and the proportion held by institutional investors, building on column (1). The ESG coefficient remained significantly positive at the 1% level, with an estimated value of 0.0310 ($t = 2.9824$, $p < 0.01$), indicating a positive impact on company value. These results suggest that ESG performance positively influences company value. Thus, hypothesis 1 is supported.

The significance and direction of the other characteristic variables are generally consistent with existing literature. The regression results show that company size and the

Table 3. Mediation effect analysis results of TFP on ESG performance and company value.

Variables	(1) Tobin Q	(2) Tobin Q	(3) TFP	(4) Tobin Q
ESG	0.0278*** (2.6398)	0.0310*** (2.9824)	0.0297*** (6.2115)	0.0260** (2.4996)
TFP				0.1679*** (6.1087)
Size	−0.6895*** (−20.9490)	−0.7804*** (−24.0230)	0.5354*** (37.3996)	−0.8703*** (−24.7341)
Age	0.6808*** (16.2026)	0.7032*** (16.6377)	0.0122 (0.6533)	0.7012*** (16.5778)
Growth	0.1107*** (6.7304)	0.0733*** (4.4911)	0.2256*** (29.3445)	0.0354** (2.1450)
Lev	0.1347 (1.2585)	0.3064*** (2.8381)	0.0317 (0.6611)	0.3011*** (2.7919)
CFO	1.0506*** (8.1296)	1.0517*** (8.4190)	0.8639*** (15.5145)	0.9066*** (7.3812)
Top1		−0.0111*** (−7.3842)	−0.0005 (−0.6693)	−0.0110*** (−7.2894)
Indep		0.0032*** (2.8788)	0.0006 (1.2722)	0.0031*** (2.7929)
INST		0.0184*** (17.9407)	0.0014*** (3.2279)	0.0182*** (17.7366)
Constant	14.6576*** (16.3285)	16.1314*** (18.9908)	−4.0425*** (−10.7283)	16.8100*** (20.1764)
Year	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Individual	Yes	Yes	Yes	Yes
N	31310	31310	31310	31310
R ²	0.2685	0.2958	0.5562	0.2988
adj. R ²	0.2664	0.2938	0.5550	0.2967

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

shareholding ratio of the largest shareholder had a significant negative impact on company value. In contrast, the results for other control variables such as company age, growth, asset-liability ratio, operating cash flow, board independence, and institutional investor shareholding indicate a significant positive impact on company value.

4.2.2. Total Factor Productivity (TFP) mediation effect tests

Table 3 presents the regression results for the mediating effect of TFP. Column (2) indicates that ESG performance had a significant total effect on company value ($a1 = 0.0310$, $p < 0.01$). Column (3) demonstrates a significant positive impact of ESG performance on TFP ($b1 = 0.0297$, $p < 0.01$). Further regression, involving both ESG performance and TFP on company value, was conducted, and the results are shown in column (4) of Table 3. TFP was found to have a significant positive effect on company value ($c2 = 0.1679$, $p < 0.01$), while ESG performance also had a significant positive impact ($c1 = 0.0260$, $p < 0.05$). Since $c1$ was less than $a1$, this indicates that TFP played a partial mediating role in the effect of ESG performance on company value. Therefore, hypothesis 2 is supported.

4.2.3. Moderation effect test for R&D innovation investment

Table 4 presents the regression results for the moderating effects of R&D innovation investment. The data used to test these effects were centered. In column (1), the coefficient of the interaction term between R&D innovation investment and ESG performance was $d3 = 0.0068$ ($t = 3.9846$, $p < 0.01$), indicating a significant positive moderating effect of R&D innovation investment on the relationship between ESG performance and company value. Specifically, higher R&D innovation investment amplified the impact of ESG performance on company value. This result suggests a 'synergistic effect' rather than a 'crowding-out effect' between R&D innovation investment and ESG performance. As illustrated in Figure 2 (left), strong R&D investment produced a large and steep slope, while weak investment resulted in a small and relatively flat slope. This finding demonstrates that greater R&D innovation investment enhances the effect of ESG performance on company value. Thus, hypothesis 3 is supported.

In column (2) of Table 4, the effect of ESG performance on TFP was analyzed. The coefficient of the interaction term between R&D innovation investment and ESG performance on TFP was $d4 = -0.0032$ ($t = -4.0280$, $p < 0.01$), indicating a negative moderating effect of R&D innovation investment on the relationship between ESG performance and TFP. This finding suggests that the influence of ESG performance on TFP diminishes as companies intensify their investments in R&D innovation. As illustrated in Figure 2 (right), the slope of the graph is steeper when R&D innovation investment is weak, but it becomes flatter with stronger R&D investment. This demonstrates that the effect of ESG performance on TFP is more pronounced when R&D innovation investment is lower. Thus, it is crucial to thoroughly examine whether R&D innovation investment enhances or diminishes this mediating effect.

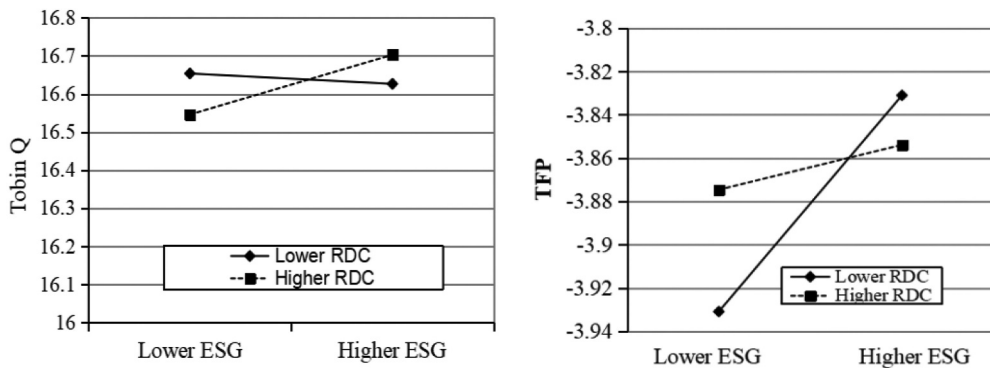
4.2.4. Moderated mediation test

In column (2) of Table 4, the regression result for the effect of ESG performance on TFP was significantly positive ($e1 = 0.0340$, $p < 0.01$), while the interaction effect between ESG performance and R&D innovation investment on TFP was significantly negative ($e3 =$

Table 4. Moderated mediation test results.

Variables	(1) Tobin Q	(2) TFP	(3) Tobin Q
ESG	0.0218** (1.9993)	0.0340*** (6.6815)	0.0181* (1.6726)
RDC	-0.0011 (-0.4664)	0.0021 (1.6160)	-0.0067*** (-2.6169)
ESG x RDC	0.0068*** (3.9846)	-0.0032*** (-4.0280)	0.0030* (1.8336)
TFP			0.2035*** (7.4101)
TFP x RDC			0.0151*** (8.3218)
Size	-0.7719*** (-24.2064)	0.5291*** (37.1497)	-0.8834*** (-25.3266)
Age	0.7008*** (16.5312)	0.0161 (0.8647)	0.6859*** (16.3056)
Growth	0.0731*** (4.4983)	0.2258*** (29.3563)	0.0360** (2.2409)
Lev	0.2934*** (2.7414)	0.0397 (0.8295)	0.2809*** (2.6780)
CFO	1.0519*** (8.4557)	0.8633*** (15.6246)	0.8770*** (7.2032)
Top1	-0.0110*** (-7.3459)	-0.0006 (-0.7500)	-0.0108*** (-7.2542)
Indep	0.0032*** (2.9257)	0.0005 (1.2037)	0.0031*** (2.8267)
INST	0.0184*** (17.9824)	0.0014*** (3.2209)	0.0182*** (17.9662)
Constant	16.0818*** (19.3668)	-3.7873*** (-9.9996)	18.6319*** (21.7500)
Year	Yes	Yes	Yes
Industry	Yes	Yes	Yes
Individual	Yes	Yes	Yes
N	31310	31310	31310
R ²	0.2971	0.5577	0.3055
adj. R ²	0.2950	0.5564	0.3034

t statistics in parentheses.* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

**Figure 2.** Moderating effect of RDC.

−0.0032, $p < 0.01$). In column (3) of Table 4, the effect of TFP on company value was significantly positive ($g1 = 0.2035$, $p < 0.01$), and the interaction term between R&D innovation investment and TFP had a significantly positive impact on company value ($g2 = 0.0151$, $p < 0.01$). The empirical results reveal that $e1$ and $g2$, $e3$ and $g1$, as well as $e3$ and $g2$, were significant across all three groups. This indicates that R&D innovation investment moderated the mediating effect of TFP in the relationship between ESG performance and company value. Therefore, hypothesis H4 is supported, suggesting that the mediating effect of TFP between ESG performance and company value is moderated by R&D innovation investment, with the conditional indirect effect being stronger under higher levels of R&D innovation investment.

4.3. Robustness and endogeneity tests

4.3.1. Robustness test

To ensure the robustness of the regression results, alternative regression analyses were conducted. Firstly, the explanatory variable Tobin's Q, which measures company value, was replaced with market value (MV) as a proxy. The regression results in column (1) of Table 5 indicate a significant positive effect of ESG on company value, with a coefficient of 0.0104 ($t = 3.2125$, $p < 0.01$). As shown in column (2) of Table 5, the inclusion of the mediating variable TFP also demonstrated a significantly positive effect, further confirming the robustness of the initial results.

Secondly, a robustness test was performed by replacing the explanatory variable. The year-end value of the China Huazheng ESG ratings was used as an alternative explanatory variable. The results in column (3) of Table 5 indicate that the regression coefficient for ESG_HZ on company value is 0.0154 ($t = 2.0244$, $p < 0.05$), demonstrating a significant positive effect. These findings align with those of previous studies, and the consistency in the direction of the key variable coefficients further reinforces the robustness of the results.

Thirdly, a robustness test was conducted by replacing the measure of the mediating variable TFP. Instead of using the LP method, TFP was measured using the OLS method

Table 5. Robustness test result.

Variables	(1) MV	(2) MV	(3) Tobin Q	(4) Tobin Q	(5) L2. Tobin Q	(6) L3. Tobin Q
ESG	0.0104*** (3.2125)	0.0081** (2.5072)		0.0256** (2.4666)	0.0418*** (3.3673)	0.0333** (2.4758)
ESG_HZ			0.0154** (2.0244)			
TFP		0.0769*** (9.4395)				
TFP_OLS				0.1863*** (6.3743)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes
Individual	Yes	Yes	Yes	Yes	Yes	Yes
N	31310	31310	31310	31310	22619	19262
R^2	0.7830	0.7850	0.2955	0.2990	0.2408	0.2768
adj. R^2	0.7823	0.7844	0.2935	0.2969	0.2379	0.2736

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

(TFP_OLS). The regression results in column (4) of Table 5 show that the coefficient for ESG is 0.0256 ($t = 2.4666$, $p < 0.05$), while the coefficient for TFP_OLS is 0.1863 ($t = 6.3743$, $p < 0.01$), both significantly positive at their respective levels. These findings further support the robustness of the regression results.

Fourthly, a robustness test was performed by lagging the explanatory variables to account for the cyclical nature of ESG performance effects and to mitigate the influence of market reaction time. The company value variables were lagged by two and three periods to complete the analysis. The results in columns (5) and (6) of Table 5 indicate that the regression coefficients for ESG remain significantly positive, further confirming the robustness of the findings.

4.3.2. Endogeneity mitigation

To address the issue of endogeneity, both lagged regression and instrumental variable regression techniques were employed.

4.3.2.1. Lagged explanatory variables. The benchmark regressions indicate a positive correlation between superior ESG performance and enhanced company value. However, the possibility of reverse causality, where higher company value could lead to improved ESG performance, cannot be ruled out. To address this issue, lagged values of ESG performance were employed, with the explanatory variable lagged by one and two periods before conducting the regressions. The results presented in column (1) of Table 6 show that the coefficient of the lagged explanatory variable (ESGt-1) is 0.0390 ($t = 3.4354$, $p < 0.01$), significant at the 1% level, suggesting robust regression outcomes. Similarly, the results in column (2) of Table 6 reveal that the coefficient for the explanatory variable lagged by two periods (ESGt-2) is 0.0295 ($t = 2.4120$, $p < 0.05$), significant at the 5% level, confirming the robustness of the regression results. These findings support the sustained positive impact of improved ESG performance on company value, aligning with previous studies and reaffirming the reliability of the research conclusions.

Table 6. Endogeneity test regression result.

Variables	(1) Tobin Q	(2) Tobin Q	(3) Phase I ESG	(4) Phase II Tobin Q
ESGt-1	0.0390*** (3.4354)			
ESGt-2		0.0295** (2.4120)		
ESG				0.0774* (1.8737)
ESG_IV			0.7961*** (30.6628)	
Controls	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Individual	Yes	Yes	Yes	Yes
N	26410	22619	31310	31310
R ²	0.2920	0.3245	0.1244	0.2948
adj. R ²	0.2897	0.3219	0.1223	0.2926

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3.2.2. Instrumental variables method. Based on the empirical results, this study employed the instrumental variable (ESG_IV) method to address endogeneity concerns. The instrumental variable used was the average ESG rating of companies within the same industry and year, which satisfied the requirements for a valid instrument as it influenced a company's ESG performance without directly affecting its value. The analysis was conducted using the two-stage least squares (2SLS) method. In the first-stage regression, a significant positive correlation was found between the instrument and ESG performance (coefficient = 0.7961, $t = 30.6628$, $p < 0.01$), confirming the validity of the instrument, with an F-statistic of 41.91, which far exceeds the weak instrument threshold of 10. In the second stage, the ESG coefficient is 0.0774 ($t = 1.8737$, $p < 0.1$), indicating a significant positive impact of ESG on company value. The adjusted R-squared values are 0.1223 in the first stage and 0.2926 in the second stage, suggesting moderate to strong explanatory power. These results demonstrate that ESG performance positively influences company value and that this relationship remains robust after addressing potential endogeneity issues.

5. Expansive research

According to Peng and Mao (2017), the industry codes for high-tech listed companies were determined based on the Guidelines for Industry Classification of Listed Companies in China (Revised 2012). In this study, the high-tech industry was defined across three sectors: manufacturing (C), information transmission, software, and information technology services (I), and scientific research and technology services (M). These sectors included 19 major categories: C25, C26, C27, C28, C29, C31, C32, C34, C35, C36, C37, C38, C39, C40, C41, I63, I64, I65, and M73. Based on this classification, the sample was divided into two groups: high-tech industry and non-high-tech industry, in order to examine the effect of ESG performance on company value, the mediation effect, and the moderating effect of R&D innovation investment, respectively. The regression results are presented in Table 6.

As shown in column (1) of Table 7, the regression coefficient for ESG is 0.0406 ($t = 3.0183$, $p < 0.01$), indicating a significant positive effect at the 1% level. Conversely, in non-high-tech industries, the regression coefficient for ESG is positive but statistically insignificant. After introducing the mediator variable TFP, the subsequent regression as presented in column (3) reveals that the ESG coefficient is 0.0347 ($t = 2.5683$, $p < 0.05$), indicating a significant positive effect at the 5% level, while ESG in non-high-tech industries remain positive but statistically insignificant. In columns (5) and (6), the moderating effect of R&D innovation investment was significantly positive at the 1% level. Notably, the regression coefficients for the moderating effect in high-tech industries were larger, suggesting a stronger positive moderation. The heterogeneity test results demonstrated that ESG performance in high-tech industries exerted a more significant and robust influence on company value compared to non-high-tech industries. This finding underscores that the impact of ESG performance on company value, as well as the moderating role of R&D innovation investment, is influenced by industry heterogeneity.

The manufacturing industry was classified according to the Guidelines for Industry Classification of Listed Companies in China (Revised 2012). The regression results are shown in Table 8. Columns (1) and (2) indicate that ESG performance had a significant

Table 7. High-tech industry heterogeneity analysis.

Variables	(1) Tobin Q High-tech Industries	(2) Tobin Q Non-high-tech Industries	(3) Tobin Q High-tech Industries	(4) Tobin Q Non-high-tech Industries	(5) Tobin Q High-tech Industries	(6) Tobin Q Non-high-tech Industries
ESG	0.0406*** (3.0183)	0.0218 (1.3735)	0.0347** (2.5683)	0.0173 (1.0854)		
TFP			0.2396*** (5.1446)	0.1047*** (2.9092)		
RDC						
ESG x RDC					0.0177*** (3.1858)	0.0047*** (2.6748)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Individual	Yes	Yes	Yes	Yes	Yes	Yes
N	18792	12518	18792	12518	18792	12518
R ²	0.3263	0.2276	0.3306	0.2295	0.3287	0.2290
adj. R ²	0.3256	0.2264	0.3299	0.2282	0.3280	0.2277

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Regression analysis variables are as above, with space constraints reporting the main variables of interest.

Table 8. Manufacturing industry heterogeneity analysis.

Variables	(1) Tobin Q Mfg	(2) Tobin Q Non-Mfg	(3) Tobin Q Mfg	(4) Tobin Q Non-Mfg	(5) TFP Mfg	(6) TFP Non-Mfg
ESG	0.0373*** (2.7938)	0.0156 (0.9309)			0.0391*** (5.1418)	0.0219** (2.1773)
TFP	0.2965*** (7.1978)	0.0449 (1.2376)				
RDC					0.0060*** (3.1459)	0.0016 (0.8549)
ESG x RDC			0.0137*** (3.3164)	0.0035* (1.9485)	−0.0023 (−1.5356)	−0.0030*** (−2.8464)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Individual	Yes	Yes	Yes	Yes	Yes	Yes
N	20797	10513	20797	10513	20797	10513
R ²	0.2973	0.2924	0.2926	0.2929	0.5878	0.4102
adj. R ²	0.2966	0.2911	0.2919	0.2914	0.5874	0.4090

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Manufacturing: Mfg, Non-manufacturing: Non-Mfg. Regression analysis variables are as above, with space constraints reporting the main variables of interest.

positive impact on the company value of the manufacturing industry, with a coefficient of 0.0373 ($t = 2.7938$, $p < 0.01$). In contrast, in the non-manufacturing industry, ESG performance had a positive but statistically insignificant impact. Columns (3) and (4) show that the interaction between ESG performance and R&D innovation investment was significant in the manufacturing industry, with a coefficient of 0.0137 ($t = 3.3164$, $p < 0.01$), indicating that R&D innovation investment positively moderated the impact of ESG performance on company value. This moderating effect was weaker in the non-manufacturing industry.

Columns (5) and (6) indicate that R&D innovation investment had a significant positive effect on TFP in the manufacturing industry (coefficient = 0.0060, $t = 3.1459$, $p < 0.01$). However, this effect was not significant in the non-manufacturing industry. In the non-manufacturing industry, R&D innovation investment negatively moderated the impact of ESG performance on TFP coefficient = −0.0030, $t = −2.8464$, $p < 0.01$). This negative moderating effect was not significant in the manufacturing industry.

Overall, these findings suggest that ESG performance and its interaction with R&D innovation investment had a stronger impact on the value of manufacturing companies compared to non-manufacturing companies, underscoring the importance of industry-specific dynamics.

6. Discussion and conclusions

6.1. Research discussion

ESG and R&D innovation represent an emerging field of study, particularly when viewed through the lens of sustainable, high-quality, and innovation-driven development. This study introduces a moderated mediation model, with TFP as a mediator and R&D innovation investment as a moderating variable, to elucidate the complex dynamics of ESG performance on company value. This framework helps guide effective resource allocation related to corporate ESG and innovation strategies.

First, the study demonstrates that TFP significantly mediates the effect of ESG performance on company value. This highlights the importance of integrating sustainable practices into corporate strategies, supporting earlier findings by L. Wang, Lian, and Dong (). ESG initiatives enhance company value both directly and indirectly by boosting TFP.

Second, in contrast to Duan, Yang, and Xiong (2023), who identified a negative moderating effect of R&D innovation investment, this study finds a synergistic impact of ESG performance and R&D innovation on company value. The differences may arise from varied research contexts. Duan et al. focused on a specific industry and a limited time-frame, whereas this study adopts a broader approach, covering multiple industries over a longer period and using total R&D investment as a more comprehensive measure of innovation capacity. By expanding both the scope and duration, this research highlights the positive synergy between ESG performance and R&D innovation investment.

The findings suggest that ESG and R&D innovation investment are not competing, but rather complementary elements of an integrated strategy that drives green innovation and achieves both environmental and innovation goals. Combining R&D with ESG initiatives enables companies to create strategic synergies, fostering stakeholder trust and enhancing competitive advantage. Strong ESG performance, paired with high R&D innovation investment, signals a commitment to sustainability and innovation, boosting investor confidence and easing financial constraints. Strategic investments in ESG and innovation promote differentiation and market success, allowing companies to develop unique, responsible products that meet consumer demand and improve valuations. This study, therefore, contributes to the literature by showing how R&D innovation investment can amplify the market impact of ESG initiatives, emphasizing the critical role of innovation in maximizing ESG-driven value creation.

Third, the positive moderating effect of R&D innovation investment is particularly evident in high-tech firms, where intensive R&D fosters better integration of innovation resources. This synergy promotes green innovation, product development, and technological advancements, enhancing both ESG contributions and company value. In high-tech sectors, strong ESG performance, coupled with robust R&D capabilities, attracts better financing conditions and stronger investor support, enabling further innovation and growth in company value.

Finally, the impact of ESG performance on company value varied significantly between manufacturing and non-manufacturing industries, underscoring the importance of industry-specific dynamics. Manufacturing companies benefited more from integrating ESG initiatives with R&D innovation investment due to their focus on resource efficiency, innovation, and technology which are factors that align closely with ESG goals and generate value-enhancing synergies. In contrast, non-manufacturing companies, which rely primarily on human capital and intangible assets, may not see immediate gains from these synergies. Therefore, these findings emphasize the need for tailored ESG strategies that account for the distinct characteristics of each sector.

6.2. Research conclusions

This study finds that ESG performance positively impacts company value, with this effect being especially strong in high-tech industries. TFP acts as a mediator, converting ESG benefits into productivity gains, which in turn enhances company value. R&D innovation investments strengthen this relationship by moderating the direct effect of ESG performance on company value and enhancing the mediating role of TFP. This effect is most pronounced in manufacturing companies, where R&D innovation investments amplify ESG benefits. These findings underscore the importance of integrating ESG initiatives with innovation strategies, particularly in maximizing company value in the manufacturing and high-tech sectors. The empirical results support the notion of ‘synergistic effects’ arising from R&D innovation investment.

6.3. Insights and recommendations

This study offers practical insights for governments, managers, and investors on how to effectively leverage the synergy between ESG performance and R&D innovation investment to enhance company value.

Governments should focus on creating targeted incentives that encourage both ESG and R&D innovation investments, such as tax breaks, green innovation subsidies, and industry-specific ESG guidelines tailored to sector characteristics. Additionally, strengthening ESG disclosure requirements and ensuring proper oversight of assessment processes are crucial for promoting transparency and accountability.

For company managers, integrating ESG initiatives with R&D strategies is key to maximizing value, especially in manufacturing and high-tech industries where TFP plays a central role. Managers should prioritize long-term strategies that focus on improving resource efficiency, technological advancement, and green innovation, rather than seeking short-term gains, to fully unlock the potential of ESG programs.

For investors, evaluating the synergy between a company's ESG initiatives and R&D innovation investments is essential. Companies that align strong ESG efforts with robust innovation capabilities are better positioned for long-term value creation, particularly in the high-tech and manufacturing sectors. These integrated approaches not only drive sustainable growth but also maximize company value and deliver higher returns across various industries.

6.4. Limitations and future research

This study provides valuable insights but also has limitations that suggest directions for future research. Firstly, the use of a single-dimensional ESG measure limits the analysis to general moderating mediation effects. Since ESG includes environmental, social, and governance dimensions, each potentially having distinct impacts, future studies should adopt multi-dimensional ESG metrics for a more nuanced analysis. Secondly, the focus on Chinese listed companies may limit the generalizability of the findings due to the unique industry characteristics and emerging market context. Future research should include samples from developed countries and conduct comparative analyses to broaden these insights across diverse industries. Thirdly, this study considers only the linear moderating effects of total R&D investment, overlooking the quality of these investments and potential non-linear impacts. Future research should explore measures of innovation quality and capacity, as well as assess both linear and non-linear moderating effects on the ESG-performance relationship. Addressing these limitations would deepen the understanding of ESG impacts in different contexts and provide further insights into the strategic integration of sustainability and R&D innovation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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